SG No. 134 D_{4h}^{12} $P4_2/nnm$ [tetragonal]

* plus set: $+[0, 0, 0]$
* Wyckoff site: `2a`, site symmetry: `-42m`

Table 1: Wyckoff bond: `2a@2a`, bond symmetry: `-42m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,2,-5,-6,-11,-12,15,16]`
2	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4,-7,-8,-9,-10,13,14]`

Table 2: Wyckoff bond: `4b@2a`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-4,7,-8]`
3	$[-X, 0, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[9,-10,-13,14]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[11,-12,15,-16]`

Table 3: Wyckoff bond: `4c@2a`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7,-8,-9,10]`

Table 4: Wyckoff bond: `8d@2a`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,-5]`
2	$[-X, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,-6]`
3	$[0, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,-8]`
4	$[0, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,-7]`
5	$[-X, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[9,-13]`
6	$[X, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[10,-14]`
7	$[0, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[11,-16]`
8	$[0, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[12,-15]`

Table 5: Wyckoff bond: `8e@2a`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,13]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,12]`
6	$[X, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6,11]`
7	$[X, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,10]`
8	$[-X, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8,9]`

Table 6: Wyckoff bond: `8f@2a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3,-4]`
3	$[-X, Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5,-6]`
4	$[Y, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15,-16]`

Table 7: Wyckoff bond: `16g@2a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2]`
3	$[-Y, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3]`
4	$[Y, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[4]`
5	$[-X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[5]`
6	$[X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8]`
9	$[-X, -Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[9]`
10	$[X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[10]`
11	$[Y, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[11]`
12	$[-Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[12]`
13	$[X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[13]`
14	$[-X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[14]`
15	$[-Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15]`
16	$[Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16]`

* Wyckoff site: `2b`, site symmetry: `-42m`

Table 8: Wyckoff bond: `2a@2b`, bond symmetry: `-42m`

No.	vector	center	mapping
1	$[0, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,2,-5,-6,-11,-12,15,16]`
2	$[0, 0, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,4,-7,-8,-9,-10,13,14]`

Table 9: Wyckoff bond: `4b@2b`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2,-5,6]`
2	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-4,7,-8]`
3	$[-X, 0, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[9,-10,-13,14]`
4	$[0, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[11,-12,15,-16]`

Table 10: Wyckoff bond: `4c@2b`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[7,-8,-9,10]`

Table 11: Wyckoff bond: `8d@2b`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-5]`
2	$[-X, 0, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,-6]`
3	$[0, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-8]`
4	$[0, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,-7]`
5	$[-X, 0, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[9,-13]`
6	$[X, 0, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[10,-14]`
7	$[0, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[11,-16]`
8	$[0, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[12,-15]`

Table 12: Wyckoff bond: `8e@2b`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,16]`
2	$[-X, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[2,15]`
3	$[-X, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,14]`
4	$[X, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[4,13]`
5	$[-X, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[5,12]`
6	$[X, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[6,11]`
7	$[X, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[7,10]`
8	$[-X, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[8,9]`

Table 13: Wyckoff bond: `8f@2b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-4]`

continued ...

Table 13

No.	vector	center	mapping
3	$[-X, Y, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[5,-6]
4	$[Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[7,-8]
5	$[-X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[9,-10]
6	$[Y, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[11,-12]
7	$[X, -Y, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[13,-14]
8	$[-Y, -X, 0]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[15,-16]

Table 14: Wyckoff bond: 16g@2b, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[1]
2	$[-X, -Y, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[2]
3	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[3]
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[4]
5	$[-X, Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[5]
6	$[X, -Y, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[6]
7	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[7]
8	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[8]
9	$[-X, -Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[9]
10	$[X, Y, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[10]
11	$[Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[11]
12	$[-Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[12]
13	$[X, -Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[13]
14	$[-X, Y, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	[14]
15	$[-Y, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[15]
16	$[Y, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	[16]

* Wyckoff site: 4c, site symmetry: 222.

Table 15: Wyckoff bond: 4a@4c, bond symmetry: 222.

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-2,-5,6]
2	$[0, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[3,-4,7,-8]
3	$[-X, 0, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[9,-10,-13,14]
4	$[0, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[11,-12,15,-16]

Table 16: Wyckoff bond: 4b@4c, bond symmetry: 222.

No.	vector	center	mapping
1	$[0, X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	[1,-2,5,-6]
2	$[-X, 0, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	[3,-4,-7,8]
3	$[0, -X, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	[9,-10,13,-14]
4	$[X, 0, 0]$	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	[11,-12,-15,16]

Table 17: Wyckoff bond: `4c@4c`, bond symmetry: `222.`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[1,2,-5,-6]`
2	$[0,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[3,4,-7,-8]`
3	$[0,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[9,10,-13,-14]`
4	$[0,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[11,12,-15,-16]`

Table 18: Wyckoff bond: `8d@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[1,-5]`
2	$[-X,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[2,-6]`
3	$[0,\ X,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[3,-8]`
4	$[0,\ -X,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[4,-7]`
5	$[-X,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[9,-13]`
6	$[X,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[10,-14]`
7	$[0,\ -X,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[11,-16]`
8	$[0,\ X,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[12,-15]`

Table 19: Wyckoff bond: `8e@4c`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ X,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[1,-6]`
2	$[0,\ -X,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[2,-5]`
3	$[-X,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[3,-7]`
4	$[X,\ 0,\ Z]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[4,-8]`
5	$[0,\ -X,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[9,-14]`
6	$[0,\ X,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[10,-13]`
7	$[X,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[11,-15]`
8	$[-X,\ 0,\ -Z]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[12,-16]`

Table 20: Wyckoff bond: `8f@4c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\ Y,\ 0]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[1,-2]`
2	$[-Y,\ X,\ 0]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[3,-4]`
3	$[-X,\ Y,\ 0]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}]$	`[5,-6]`
4	$[Y,\ X,\ 0]$	$[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}]$	`[7,-8]`
5	$[-X,\ -Y,\ 0]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[9,-10]`
6	$[Y,\ -X,\ 0]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[11,-12]`
7	$[X,\ -Y,\ 0]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}]$	`[13,-14]`
8	$[-Y,\ -X,\ 0]$	$[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}]$	`[15,-16]`

Table 21: Wyckoff bond: `16g@4c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[16]`

* Wyckoff site: `4d`, site symmetry: `2.22`

Table 22: Wyckoff bond: `4a@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,-12,13,-14]`

Table 23: Wyckoff bond: `4b@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-2,-7,8]`
2	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,-4,-5,6]`
3	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,-10,-15,16]`
4	$[-X, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,-12,-13,14]`

Table 24: Wyckoff bond: `4c@4d`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,2,-7,-8]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,4,-5,-6]`
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,10,-15,-16]`
4	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,12,-13,-14]`

Table 25: Wyckoff bond: `8d@4d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-8]`
2	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2,-7]`
3	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,-6]`
4	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,-5]`
5	$[-X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,-13]`

Table 26: Wyckoff bond: `8e@4d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,-14]`

Table 27: Wyckoff bond: `8f@4d`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[15,-16]`

Table 28: Wyckoff bond: `16g@4d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[7]`

continued ...

Table 28

No.	vector	center	mapping
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[16]`

* Wyckoff site: `4e`, site symmetry: `..2/m`

Table 29: Wyckoff bond: `4a@4e`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-7,-9,15]`
2	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-8,-10,16]`
3	$[X, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-5,-11,13]`
4	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-6,-12,14]`

Table 30: Wyckoff bond: `4b@4e`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,7,-9,-15]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,8,-10,-16]`
3	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,5,-11,-13]`
4	$[X, -X, 0]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,6,-12,-14]`

Table 31: Wyckoff bond: `8c@4e`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, 0\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, 0\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$\left[0, 0, \frac{1}{2}\right]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[8,-16]`

* Wyckoff site: `4f`, site symmetry: `..2/m`

Table 32: Wyckoff bond: `4a@4f`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[0, 0, 0]$	`[1,-7,-9,15]`
2	$[-X, X, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-8,-10,16]`
3	$[X, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-5,-11,13]`
4	$[-X, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-6,-12,14]`

Table 33: Wyckoff bond: `4b@4f`, bond symmetry: `..2/m`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, 0, 0]$	`[1,7,-9,-15]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,8,-10,-16]`
3	$[-X, X, 0]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,5,-11,-13]`
4	$[X, -X, 0]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,6,-12,-14]`

Table 34: Wyckoff bond: `8c@4f`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,-10]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,-11]`
4	$[Y, -X, Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[4,-12]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,-13]`
6	$[X, -Y, -Z]$	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[6,-14]`
7	$[Y, X, -Z]$	$[0, 0, 0]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[8,-16]`

* Wyckoff site: `4g`, site symmetry: `2.mm`

Table 35: Wyckoff bond: `4a@4g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,-2,-15,16]`
2	$[-X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,-4,-13,14]`
3	$[-X, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,-6,-11,12]`
4	$[X, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,-8,-9,10]`

Table 36: Wyckoff bond: `4b@4g`, bond symmetry: `2.mm`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,2,15,16]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,4,13,14]`
3	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$[0, 0, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,10]`

Table 37: Wyckoff bond: `8c@4g`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,16]`
2	$[-X, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[2,15]`
3	$[-X, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,14]`
4	$[X, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[4,13]`
5	$[-X, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,12]`
6	$[X, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[6,11]`
7	$[X, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,10]`
8	$[-X, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[8,9]`

Table 38: Wyckoff bond: `8d@4g`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[15,-16]`

Table 39: Wyckoff bond: `16e@4g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[16]`

* Wyckoff site: `8h`, site symmetry: `2..`

Table 40: Wyckoff bond: `8a@8h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[15,-16]`

Table 41: Wyckoff bond: `8b@8h`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,2]`
2	$[0, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3,4]`
3	$[0, 0, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,6]`
4	$[0, 0, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[7,8]`
5	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9,10]`
6	$[0, 0, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11,12]`
7	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13,14]`
8	$[0, 0, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[15,16]`

Table 42: Wyckoff bond: `16c@8h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[16]`

* Wyckoff site: `8i`, site symmetry: `.2.`

Table 43: Wyckoff bond: `8a@8i`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ X,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,-6]`
2	$[0,\ -X,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,-5]`
3	$[-X,\ 0,\ Z]$	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[3,-7]`
4	$[X,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ -Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,-14]`
6	$[0,\ X,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,-13]`
7	$[X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[11,-15]`
8	$[-X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[12,-16]`

Table 44: Wyckoff bond: `8b@8i`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1,6]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2,5]`
3	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[3,7]`
4	$[0,\ -X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4,8]`
5	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9,14]`
6	$[X,\ 0,\ 0]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10,13]`
7	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[11,15]`
8	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[12,16]`

Table 45: Wyckoff bond: `16c@8i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	`[16]`

* Wyckoff site: `8j`, site symmetry: `.2.`

Table 46: Wyckoff bond: `8a@8j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0,\ X,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,-6]`
2	$[0,\ -X,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,-5]`
3	$[-X,\ 0,\ Z]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[3,-7]`
4	$[X,\ 0,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[4,-8]`
5	$[0,\ -X,\ -Z]$	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9,-14]`
6	$[0,\ X,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[10,-13]`
7	$[X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[11,-15]`
8	$[-X,\ 0,\ -Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[12,-16]`

Table 47: Wyckoff bond: `8b@8j`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,6]`
2	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,5]`
3	$[0,\ X,\ 0]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[3,7]`
4	$[0,\ -X,\ 0]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[4,8]`
5	$[-X,\ 0,\ 0]$	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9,14]`
6	$[X,\ 0,\ 0]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[10,13]`
7	$[0,\ -X,\ 0]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[11,15]`
8	$[0,\ X,\ 0]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[12,16]`

Table 48: Wyckoff bond: `16c@8j`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[16]`

* Wyckoff site: `8k`, site symmetry: `..2`

Table 49: Wyckoff bond: `8a@8k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, x, 0]$	`[1,-7]`
2	$[-X, X, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, 0\right]$	`[2,-8]`
3	$[X, X, Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}\right]$	`[3,-5]`
4	$[-X, -X, Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4,-6]`
5	$[-X, X, -Z]$	$[-x, -x, 0]$	`[9,-15]`
6	$[X, -X, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, 0\right]$	`[10,-16]`
7	$[-X, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}\right]$	`[11,-13]`
8	$[X, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12,-14]`

Table 50: Wyckoff bond: `8b@8k`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, x, 0]$	`[1,7]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, 0\right]$	`[2,8]`
3	$[-X, X, 0]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}\right]$	`[3,5]`
4	$[X, -X, 0]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4,6]`
5	$[-X, -X, 0]$	$[-x, -x, 0]$	`[9,15]`
6	$[X, X, 0]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, 0\right]$	`[10,16]`
7	$[X, -X, 0]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}\right]$	`[11,13]`
8	$[-X, X, 0]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12,14]`

Table 51: Wyckoff bond: `16c@8k`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, 0\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, x, \frac{1}{2}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, \frac{1}{2}-x, \frac{1}{2}\right]$	`[6]`
7	$[Y, X, -Z]$	$[x, x, 0]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-x, \frac{1}{2}-x, 0\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -x, 0]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, 0\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -x, \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, x+\frac{1}{2}, \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$[-x, -x, 0]$	`[15]`
16	$[Y, X, Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{2}, 0\right]$	`[16]`

* Wyckoff site: `8l`, site symmetry: `..2`

Table 52: Wyckoff bond: `8a@8l`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\,-X,\,Z]$	$\left[x,\,x,\,\frac{1}{2}\right]$	`[1,-7]`
2	$[-X,\,X,\,Z]$	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$	`[2,-8]`
3	$[X,\,X,\,Z]$	$\left[\frac{1}{2}-x,\,x,\,0\right]$	`[3,-5]`
4	$[-X,\,-X,\,Z]$	$\left[x,\,\frac{1}{2}-x,\,0\right]$	`[4,-6]`
5	$[-X,\,X,\,-Z]$	$\left[-x,\,-x,\,\frac{1}{2}\right]$	`[9,-15]`
6	$[X,\,-X,\,-Z]$	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}\right]$	`[10,-16]`
7	$[-X,\,-X,\,-Z]$	$\left[x+\frac{1}{2},\,-x,\,0\right]$	`[11,-13]`
8	$[X,\,X,\,-Z]$	$\left[-x,\,x+\frac{1}{2},\,0\right]$	`[12,-14]`

Table 53: Wyckoff bond: `8b@8l`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\,X,\,0]$	$\left[x,\,x,\,\frac{1}{2}\right]$	`[1,7]`
2	$[-X,\,-X,\,0]$	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$	`[2,8]`
3	$[-X,\,X,\,0]$	$\left[\frac{1}{2}-x,\,x,\,0\right]$	`[3,5]`
4	$[X,\,-X,\,0]$	$\left[x,\,\frac{1}{2}-x,\,0\right]$	`[4,6]`
5	$[-X,\,-X,\,0]$	$\left[-x,\,-x,\,\frac{1}{2}\right]$	`[9,15]`
6	$[X,\,X,\,0]$	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}\right]$	`[10,16]`
7	$[X,\,-X,\,0]$	$\left[x+\frac{1}{2},\,-x,\,0\right]$	`[11,13]`
8	$[-X,\,X,\,0]$	$\left[-x,\,x+\frac{1}{2},\,0\right]$	`[12,14]`

Table 54: Wyckoff bond: `16c@8l`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[x,\,x,\,\frac{1}{2}\right]$	`[1]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$	`[2]`
3	$[-Y,\,X,\,Z]$	$\left[\frac{1}{2}-x,\,x,\,0\right]$	`[3]`
4	$[Y,\,-X,\,Z]$	$\left[x,\,\frac{1}{2}-x,\,0\right]$	`[4]`
5	$[-X,\,Y,\,-Z]$	$\left[\frac{1}{2}-x,\,x,\,0\right]$	`[5]`
6	$[X,\,-Y,\,-Z]$	$\left[x,\,\frac{1}{2}-x,\,0\right]$	`[6]`
7	$[Y,\,X,\,-Z]$	$\left[x,\,x,\,\frac{1}{2}\right]$	`[7]`
8	$[-Y,\,-X,\,-Z]$	$\left[\frac{1}{2}-x,\,\frac{1}{2}-x,\,\frac{1}{2}\right]$	`[8]`
9	$[-X,\,-Y,\,-Z]$	$\left[-x,\,-x,\,\frac{1}{2}\right]$	`[9]`
10	$[X,\,Y,\,-Z]$	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}\right]$	`[10]`
11	$[Y,\,-X,\,-Z]$	$\left[x+\frac{1}{2},\,-x,\,0\right]$	`[11]`
12	$[-Y,\,X,\,-Z]$	$\left[-x,\,x+\frac{1}{2},\,0\right]$	`[12]`
13	$[X,\,-Y,\,Z]$	$\left[x+\frac{1}{2},\,-x,\,0\right]$	`[13]`
14	$[-X,\,Y,\,Z]$	$\left[-x,\,x+\frac{1}{2},\,0\right]$	`[14]`
15	$[-Y,\,-X,\,Z]$	$\left[-x,\,-x,\,\frac{1}{2}\right]$	`[15]`
16	$[Y,\,X,\,Z]$	$\left[x+\frac{1}{2},\,x+\frac{1}{2},\,\frac{1}{2}\right]$	`[16]`

* Wyckoff site: `8m`, site symmetry: `..m`

Table 55: Wyckoff bond: `8a@8m`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, -x, z]$	`[1,15]`
2	$[-X, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, z\right]$	`[2,16]`
3	$[X, X, Z]$	$\left[x + \frac{1}{2}, x, z + \frac{1}{2}\right]$	`[3,13]`
4	$[-X, -X, Z]$	$\left[-x, \frac{1}{2} - x, z + \frac{1}{2}\right]$	`[4,14]`
5	$[-X, -X, -Z]$	$\left[\frac{1}{2} - x, -x, \frac{1}{2} - z\right]$	`[5,11]`
6	$[X, X, -Z]$	$\left[x, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[6,12]`
7	$[-X, X, -Z]$	$[-x, x, -z]$	`[7,9]`
8	$[X, -X, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[8,10]`

Table 56: Wyckoff bond: `8b@8m`, bond symmetry: `..m`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, -x, z]$	`[1,-15]`
2	$[-X, -X, 0]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, z\right]$	`[2,-16]`
3	$[-X, X, 0]$	$\left[x + \frac{1}{2}, x, z + \frac{1}{2}\right]$	`[3,-13]`
4	$[X, -X, 0]$	$\left[-x, \frac{1}{2} - x, z + \frac{1}{2}\right]$	`[4,-14]`
5	$[-X, X, 0]$	$\left[\frac{1}{2} - x, -x, \frac{1}{2} - z\right]$	`[5,-11]`
6	$[X, -X, 0]$	$\left[x, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[6,-12]`
7	$[X, X, 0]$	$[-x, x, -z]$	`[7,-9]`
8	$[-X, -X, 0]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[8,-10]`

Table 57: Wyckoff bond: `16c@8m`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, -x, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, z\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[x + \frac{1}{2}, x, z + \frac{1}{2}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[-x, \frac{1}{2} - x, z + \frac{1}{2}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, -x, \frac{1}{2} - z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[6]`
7	$[Y, X, -Z]$	$[-x, x, -z]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, x, -z]$	`[9]`
10	$[X, Y, -Z]$	$\left[x + \frac{1}{2}, \frac{1}{2} - x, -z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2} - x, -x, \frac{1}{2} - z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[x, x + \frac{1}{2}, \frac{1}{2} - z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x + \frac{1}{2}, x, z + \frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[-x, \frac{1}{2} - x, z + \frac{1}{2}\right]$	`[14]`
15	$[-Y, -X, Z]$	$[x, -x, z]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, z\right]$	`[16]`

* Wyckoff site: `16n`, site symmetry: `1`

Table 58: Wyckoff bond: `16a@16n`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[2]`
3	$[-Y,\ X,\ Z]$	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[3]`
4	$[Y,\ -X,\ Z]$	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[4]`
5	$[-X,\ Y,\ -Z]$	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[5]`
6	$[X,\ -Y,\ -Z]$	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[Y,\ X,\ -Z]$	$[y,\ x,\ -z]$	`[7]`
8	$[-Y,\ -X,\ -Z]$	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[8]`
9	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[9]`
10	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[10]`
11	$[Y,\ -X,\ -Z]$	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[11]`
12	$[-Y,\ X,\ -Z]$	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[12]`
13	$[X,\ -Y,\ Z]$	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[13]`
14	$[-X,\ Y,\ Z]$	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[14]`
15	$[-Y,\ -X,\ Z]$	$[-y,\ -x,\ z]$	`[15]`
16	$[Y,\ X,\ Z]$	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16]`